

JANUS Optical Head Line of Sight Temperature dependence Characterization and Validation by on ground test

Andrea Turella

Leonardo S.p.A.

Campi Bisenzio, Italy

andrea.turella@leonardocompany.com

Vincenzo Della Corte

INAF-IAPS

Rome, Italy

vincenzo.dellacorte@inaf.it

Pasquale Palumbo

Parthenope University

Napoli, Italy

pasquale.palumbo@uniparthenope.it

Marilena Amoroso

ASI

Matera, Italy

marilena.amoroso@asi.it

Raffaele Mugnuolo

ASI

Matera, Italy

raffaele.mugnuolo@asi.it

Giovanni Enrico Noci

Leonardo S.p.A.

Campi Bisenzio, Italy

giovannienrico.noci@leonardocompany.com

Abstract— JANUS (Jupiter Amorum ac Natorum Undique Scrutator) is the multispectral camera that, on board of JUICE (JUPiter ICy moons Explorer) ESA mission, will provide high-resolution images of Jupiter atmosphere and major Galilean icy moons surface: Europa, Ganymede, Callisto. Inertial pointing, mosaicking, object tracking, stereo imaging for DTM (Digital Terrain Model) and the peculiar Ganymede Libration measurements defined highly stringent requirement on the LoS (Line of Sight) knowledge of the JANUS OHU (Optical Head Unit). Once in flight, the JANUS LoS is expected to vary as consequence of thermal environment variations, in fact, orbit attitude change, spacecraft maneuvering and scientific payloads heat dissipations can modify the OHU temperature distribution. Optical elements translation and rotation have been predicted through a STOP (Structural Thermal Optical and Performance) Analysis, deriving a linear relation between LoS, value and uncertainty, and OHU structure temperature gradient. This relation has been validated thorough an on-ground and in-air characterization test, the results are presented in this paper. Test based coefficients allow real time and indirect LoS prediction & evaluation as function of OHU temperature sensor readings.

Keywords— JUICE ESA mission, JANUS, Optical stability, Performances, Line of Sight Space Instrument, High resolution camera

I. INTRODUCTION

The JUUpiter ICy moons Explorer (JUICE) mission will survey the Jovian system with a special focus on the three Galilean Moons – Europa, Ganymede and Callisto – becoming the first SpaceCraft (S/C) to orbit around the Jupiter moon Ganymede [1]. The main objective of JUICE is to address two key questions: what are the conditions for planet formation and the emergence of life? And how does the Solar System work? The main components of the mission scientific strategy are: 1) to conduct a comparative study of Ganymede, Callisto and Europa 2) to provide a complete spatial-temporal characterization of the giant, rotating magnetosphere, and of the meteorology, chemistry and

structure of Jupiter's gaseous atmosphere, and 3) to study coupling processes inside Jupiter's system and in particular about the two key coupling processes within that system: the tidal effects that couple Jupiter with its satellites, and the electrodynamic interactions that couple Jupiter and its satellites with their atmospheres, subsurface oceans, magnetospheres and magnetodisc.

JANUS (Jovis Amorum ac Natorum Undique Scrutator) is the narrow-angle camera selected as the visible (from near UV to NIR) imager onboard JUICE. The JANUS instrument is composed by three units:

- Optical Head Unit (OHU), mounted on the S/C Optical Bench;
- Proximity Electronics Unit (PEU);
- Main Electronics Unit (MEU), located in the S/C vault.

To meet the mission scientific requirements related to the Jovian Atmosphere investigations and Ganymede Libration measurements, the knowledge and characterization of the Line Of Sight (LOS) of the OHU is crucial. In [2] a Structural-Thermal-Optical-Performance (STOP) analysis of the prediction of the LOS behaviour has been performed. In this paper the verification by on ground test measurements of this analysis is reported.

II. LOS CHARACTERIZATION WITH IN-AIR TEST

The test focused on the main contributor of the LoS, that is the OMS (Opto Mechanical Structure) [3] deformation due to BW (Baffle Wall) and OW (Optical Wall) temperature gradient.

According to [2], Y axis rotation is dominant wrt (with respect to) X rotation, for this reason:

$$\text{LoS} = \text{LoS}_Y \quad (1)$$

Knowing that all other sources of LoS variation are of negligible effect, the test was conducted using the OHU STM (Structural Thermal Model), including FPM (Focal Plane Module) STM and FWM (Filter Wheel Module) STM. External harness and the S/C purging line were not installed.

The involvement of Leonardo as JANUS Instrument Prime contractor and OHU design authority is performed in the frame of Italian Space Agency (ASI) Industrial Contract N. 2018-01-I.O. The Italian Principal Investigator team acknowledges support from ASI under ASI-INAF agreement N. 2018-25-HH.0.

Contributors to LoS such like FPM Detector / PCB deformation and FWM Filters rotation/deformation were not considered.

A. Concept

Installed on an aluminum rigid interface plate (kept at ambient temperature) the OW was heated to obtain a temperature gradient wrt the BW. This gradient led to a differential contraction in X direction of the OW wrt BW, indeed causing a flexure of the full OMS (i.e. Y rotation).

The temperature difference between OW and BW is named: Wall's temperature gradient (ΔT_{Walls}).

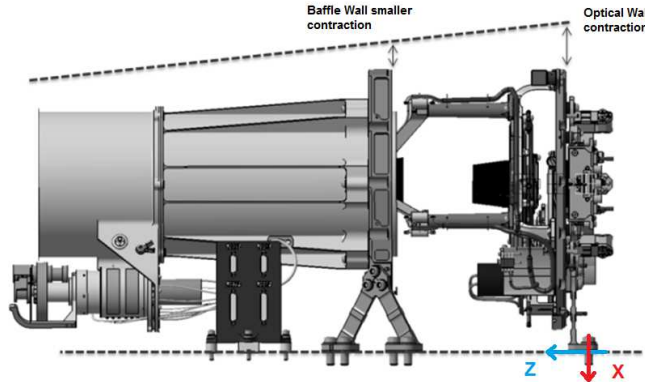


Fig. 1. : OHU LoS variation due to Walls temperature gradient (spatial)

B. Assumption

- Primary Mirror tilt = LoS of OHU

The LoS main contribution is the global tilting of OHU structure. Being TEM (Telescope Module) stiffly connected to the OW, any rotation of the OW causes TEM optical components rotation. So, the measurement of the M1 (Primary Mirror) rotation is equivalent to measure the whole TEM rotation, and of all OHU LoS variation.

- Air environment and convection effects

Involving only the reflection for theodolite measurement, the test was done in air.

The advantages of testing in air were: a) the constant accessibility to the OHU and the setup; b) the short transitory of the temperature distribution accelerated thanks to convection; c) the absence of «chamber windows bending» due to vacuum that affect the theodolite measurement uncertainty (the LoS variation was previously tested in vacuum, but the depressurized condition induced the window to bend; such bending introduced uncertainty during measurement of more than 60 arcsec, larger than the measure itself).

The disadvantage were: a) the convection constantly chilling the OHU so heating power to be injected was widely lost (i.e. setup required high Wattage); b) convection enhanced the hot spots location due to heating injection and reduced the uniformity of the temperature distribution.

- Temperature stability in air similar to in flight

The test data were collected when the variation of the M1 temperature wrt the OW temperature was slower

than 2°C/h. From OHU in flight thermal analysis (Ganymede 500km orbit), the variation is predicted to be 1°C/h.

So, the collection of data during in air-lab test was done when thermal transient was comparable and any eventual different elongation, that could have induced a relative rotation between M1 and OW, was considered negligible wrt global OMS angular rotation (confirmed by analysis [2]).

- Heating source location representativeness

Test heaters installed were sufficiently representative of the hot spots expected from thermal analysis, see Fig. 4.

- Interface

OHU interface was rigidly fixed on an aluminum plate kept at room temperature to avoid distortion due to differential thermo-elastic elongation. The plate needed to be monitored to check its constant temperature all along the test. No external harnesses were installed.

- Theodolite measurement

The rotation of the M1 was measured with a Theodolite, stable and permanently fixed on the lab floor. Since STM flat mirror was uncoated, an Optical Cube was glued on its surface, see Fig. 3.

C. Setup

The adopted test setup is shown in Figs. 2, 3 and 4.

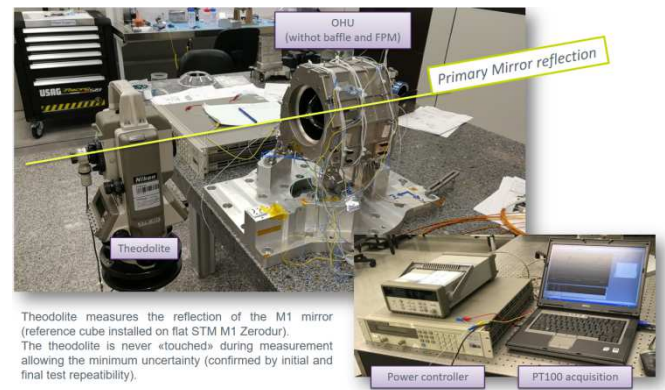
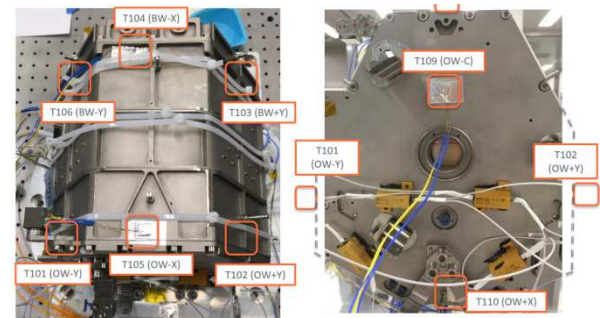


Fig. 2. : Test Setup chain

PT100 were used to measure the OHU temperature in its main location, and to derive the average temperature of the elements: OW, BW, M1 and Aluminum interface plate.



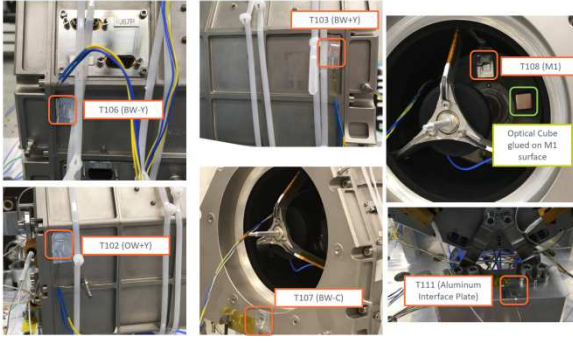


Fig. 3. : Test Setup PT100 locations and Optical Cube on M1

Dedicated heaters of 50 Watts were placed in the lower side of the OW to reproduce the hot spot locations showed in OHU thermal analysis [2] (once in orbit, the heat on the OHU will be mainly conducted by harnesses).

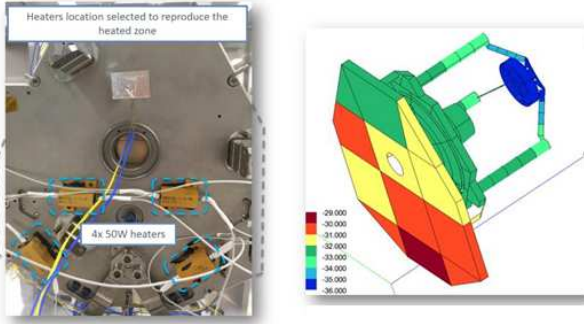


Fig. 4. : Test Setup Heaters location and expected in-flight hot spot locations from thermal analysis

D. Test Flow

The heating power was increased up to the limit of the power supply (85 Watts) to generate the maximum gradient between Walls, with the objective of enhancing the deformation of the OMS for a less impacting theodolite measurement uncertainty.

All theodolite measurements were taken when the temperature difference of M1 (which is the slowest temperature varying element during the transient and therefore can be identified as the higher thermal resistance path) and OW was stabilized within 2°C/h.

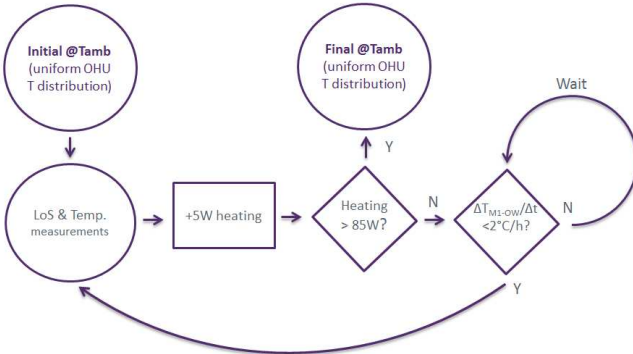


Fig. 5. : Test flow.

E. Test Data

The test data collected are presented in graphical form in Figs. 6 and 7. Heat power is shown by black arrows, each of them representing 5 Watts step increase. Note that the

rotation around X is confirmed to be negligible wrt Y rotation.

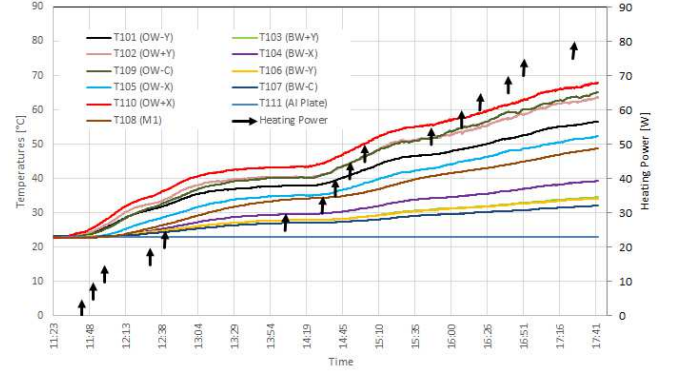


Fig. 6. : Test data: Temperature measurements and injected Heating

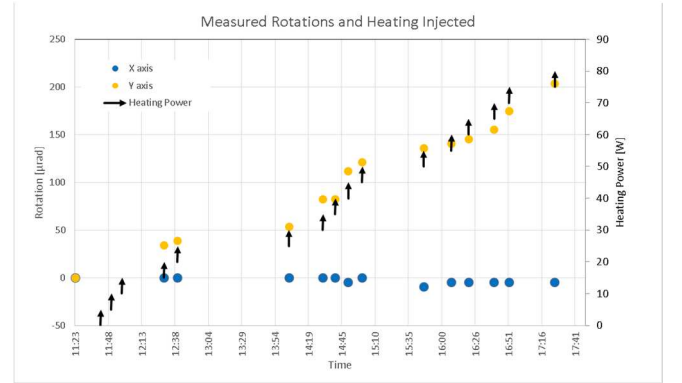


Fig. 7. : Test data: Rotation measurements and Heating injected

III. 6 LOS PROPORTIONALITY FACTOR

The proportionality factor, according to [2], is defined as:

$$K = \text{LoS} / \Delta T_{\text{walls}} \pm u_K \quad (2)$$

A. Assumptions

For the calculation of the gradient some assumptions are made.

- The temperature of the OW and BW are equal to the average of all thermistors measurements available.

$$T_{OW} = \text{avg} (T_{OW_i}) \quad (3)$$

$$T_{BW} = \text{avg} (T_{BW_j}) \quad (4)$$

- The uncertainty of the Walls temperatures, due to the position of the sensors and the non homogeneity of the surface temperatures (caused by the heaters plus convection that generated the hot spots) will be determined taking into account the dispersion of the values measured:

$$uT_{OW} = \pm \sqrt{\frac{1}{i-1} \sum_i (T_{OW_i} - \text{avg}(T_{OW}))^2} \quad (5)$$

$$uT_{BW} = \pm \sqrt{\frac{1}{j-1} \sum_j (T_{BW_j} - \text{avg}(T_{BW}))^2} \quad (6)$$

- The uncertainty is also variable as function of the heating power. Non homogeneity increases when heating power increases. The larger uncertainty is always associated to OW (e.g. at 0 Watt, $uT_{OW} = uT_{BW} = \pm 0^\circ\text{C}$, at 85W $uT_{OW} = \pm 3.3^\circ\text{C}$ and $uT_{BW} = \pm 1^\circ\text{C}$). So, applying Kline McClintock formulation [4], the gradient uncertainty can be approximated as:

$$u\Delta T_{Walls} = \sqrt{uT_{OW}^2 + uT_{BW}^2} \sim uT_{OW} \quad (7)$$

- The PT100 measurement uncertainty is $\pm 0.5^\circ\text{C}$ and it is considered negligible.
- The theodolite uncertainty is assumed to be (based on technical laboratory experience): $uLoS = 5\text{arcsec} = 24\mu\text{rad}$.

B. Proportionality Factor

Adopting the proportional relation between LoS and ΔT_{Walls} , in agreement with the multidisciplinary study in (2), the K proportional factor was calculated as best linear fit of the test data. The fit was imposed to pass for the reference measurement at ambient temperature, where by definition the LoS is zero when no temperature gradient is present on the structure. And the uncertainty calculation of the K factor is obtained as well as linear fit of the measurements including uncertainty values.

The fit shows a partial agreement of the test measurements at low ΔT_{Walls} , the causes is probably the not completion of the transitory (i.e. the temperature of the structure was rising slowly, not stabilized yet, causing low distortion being still close to ambient temperature).

Fig. 8 shows also the LoS predicted with the STOP FEM (Finite Element Model) [2] in which the temperature distribution measured during the test (at 85 W) was imposed as analysis input. The FEM model was updated with the CTE measurement ($2.0 \cdot 10^{-6} \text{ 1/K}$) of the Invar36 provided by the manufacturer – OMS bulk material. The FEM predicted LoS proportional factor K is in agreement with the uncertainty region of the test results.

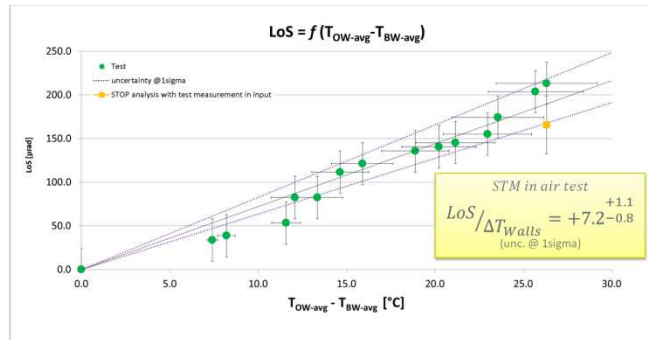


Fig. 8. : LoS vs Walls gradient temperature, analysis prediction, test results, uncertainties and best linear fit

IV. 6 LoS & AKE IN-FLIGHT PREDICTION

The LoS can be expressed as:

$$LoS(t) = K \cdot \Delta T_{Walls}(t) \pm u_{LoS}(t) \quad (8)$$

The AKE (Absolute Knowledge Error of Pointing) is defined as the instantaneous knowledge of the pointing error

(difference between the actual spacecraft attitude and the desired target attitude guidance reference) at any given time. Part of the AKE is caused by the S/C attitude control and S/C structural deformations; the remaining part is caused by the variation of the JANUS LoS. The LoS reference is determined during star calibration (sky pointing). Any variation from the reference LoS during the scientific acquisition around Ganymede is.

$$AKE(t) = LoS(Science) - LoS(Calibration) = K \cdot (\Delta T_{Walls}^{Science}(t) - \Delta T_{Walls}^{Calibration}) \pm u_{AKE}(t) \quad (9)$$

where t is the instantaneous time in which Walls gradient is measured. Please note that ΔT_{Walls} during calibration is not time dependent being a single calibration activity.

A. Walls gradient measurement during flight

When OHU will be in space, during imaging while orbiting, the measurements of the Wall's temperature will be recorded by two PT1000 in specific locations of the box, see Fig. 9.

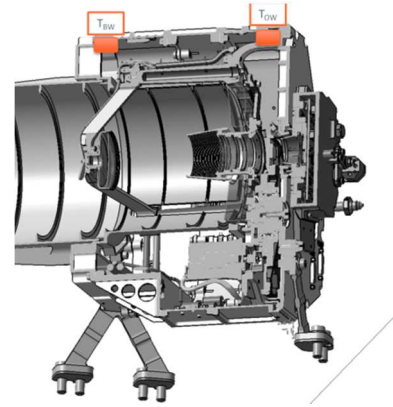


Fig. 9. : Location of PT1000 in OHU Proto Flight Model

To estimate the uncertainty of the Walls gradient measurement using only two sensors (i.e. the temperature measurement might be representative only of the local position), OHU STM TVC (Thermal Vacuum Cycling) test data were port-processed. A wide set of PT100 sensors (at different Walls locations) were registered, offering a direct evaluation of the Walls temperature homogeneity. During TVC, thanks to the absence of convection and the slow boundary condition change, the gradient based on sensors BW-X and OW-X (see Fig. 9 and 10 to appreciate the identical location of PT100 and PT1000) resulted very close to the value estimated by averaging all available sensors.

In Fig. 11, the difference between the gradient measurement methods (averaging of all sensors available versus use of only two, like on PFM) is represented by a green line. The maximum and the minimum peak occurred in correspondence of the STM vacuum test boundaries switch (fast change from/to hot/cold non operative environments). These unrealistic variation will never be seen in flight (slower time scale of Spacecraft Attitude and Orbit changes during JANUS imaging) High temperature uniformity of OW and BW are expected. The uncertainty is assumed at 1-sigma as the standard deviation over the whole temperature history recorded in STM TVC test.

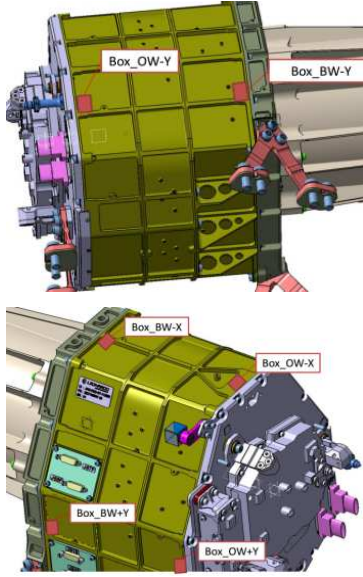


Fig. 10. : Location of PT1000 in OHU Structural Thermal Model during Thermal Vacuum Cycling test

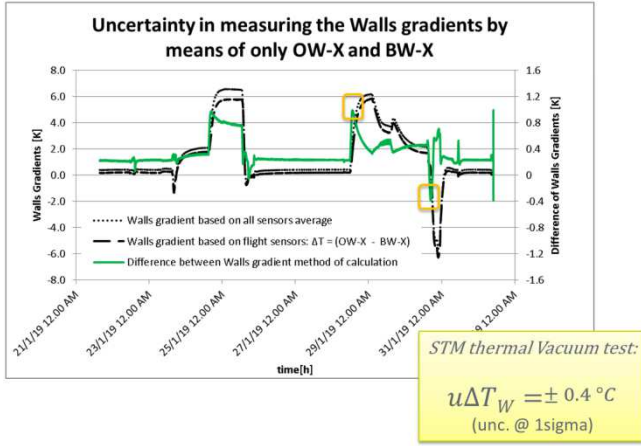


Fig. 11. : Uncertainty in measuring the Walls gradient by means of only two PT1000, data from OHU STM thermal vacuum cycling test

B. LoS empirical functions

Neglecting the uncertainty due to the PT1000 measurement (i.e. $\pm 0.1^\circ\text{C}$ from datasheet) and knowing that:

$$uLoS(t) = \sqrt{(\Delta T_{Walls}(t) \cdot u_K)^2 + (K \cdot u_{\Delta T_{Walls}})^2} \quad (10)$$

the graphical and tabular form of the LoS function of the Walls gradient is presented in Fig. 12 and 13 with associated 2 sigma uncertainty (as to mission scientific requirement). It is assumed that inverting the Walls gradient, also the LoS and AKE change sign maintaining the proportional relation.

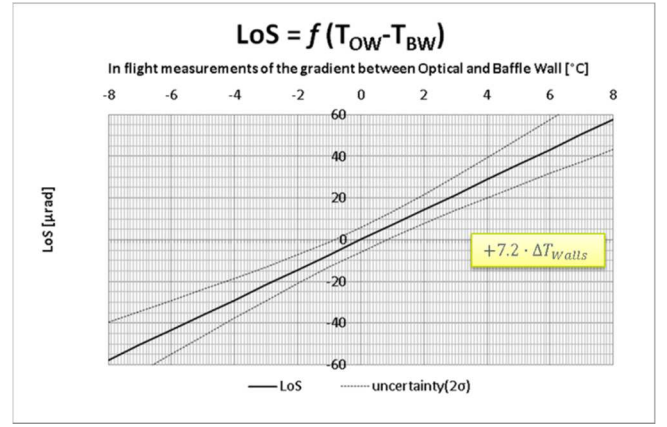


Fig. 12. : Graphical form of LoS function of Wall's gradient (2 sigma uncertainty)

OW _{PT1000} - BW _{PT1000}	LoS	+unc 2σ	-unc 2σ
[°C]	[μrad]	[μrad]	[μrad]
-8	-58	18	-15
-7	-50	16	-13
-6	-43	14	-12
-5	-36	12	-10
-4	-29	10	-9
-3	-22	9	-8
-2	-14	7	-7
-1	-7	6	-6
0	0	6	-6
1	7	6	-6
2	14	7	-7
3	22	9	-8
4	29	10	-9
5	36	12	-10
6	43	14	-12
7	50	16	-13
8	58	18	-15

Fig. 13. : Tabular form of LoS function of Wall's gradient (2 sigma uncertainty)

C. AKE empirical functions

Neglecting the uncertainty due to the PT1000 measurement (i.e. $\pm 0.1^\circ\text{C}$ from datasheet) and knowing that:

$$u_{AKE}(t) = \sqrt{[(\Delta T_{Walls}^{Science}(t) - \Delta T_{Walls}^{Calibration}(t)) \cdot u_K]^2 + 2 \cdot (K \cdot u_{\Delta T_{Walls}})^2} \quad (11)$$

where conservatively $u\Delta T_{Walls}^{Science} \sim u\Delta T_{Walls}^{Calibration}$, the graphical and tabular form of the AKE function of the Walls gradient variation are reported in Fig. 14 and 15. Uncertainty is expressed at 2 sigma (as to mission scientific requirement).

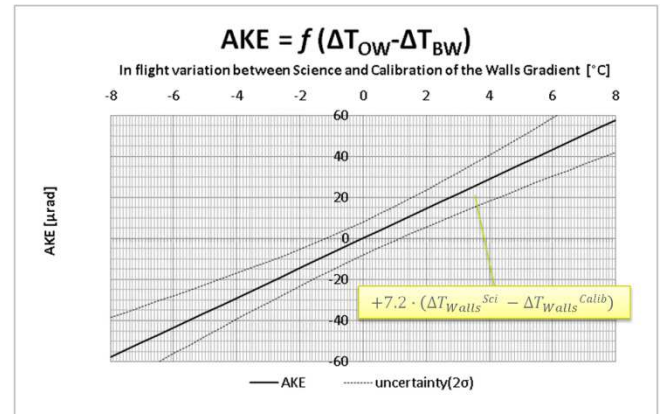


Fig. 14. : Graphical form of AKE function of Wall's gradient (2 sigma uncertainty)

$(OW_{PT1000} - BW_{PT1000})^{std} - (OW_{PT1000} - BW_{PT1000})^{calc}$	AKE	+unc 2σ	-unc 2σ
[°C]	[μrad]	[μrad]	[μrad]
-8	-58	19	-16
-7.5	-54	18	-15
-7	-50	17	-14
-6.5	-47	16	-14
-6	-43	15	-13
-5.5	-40	14	-12
-5	-36	14	-12
-4.5	-32	13	-11
-4	-29	12	-11
-3.5	-25	11	-10
-3	-22	10	-10
-2.5	-18	10	-9
-2	-14	9	-9
-1.5	-11	9	-9
-1	-7	8	-8
-0.5	-4	8	-8
0	0	8	-8
0.5	4	8	-8
1	7	8	-8
1.5	11	9	-9
2	14	9	-9
2.5	18	10	-9
3	22	10	-10
3.5	25	11	-10
4	29	12	-11
4.5	32	13	-11
5	36	14	-12
5.5	40	14	-12
6	43	15	-13
6.5	47	16	-14
7	50	17	-14
7.5	54	18	-15
8	58	19	-16

Fig. 15. : Tabular form of AKE function of Wall's gradient (2 sigma uncertainty)

V. CONCLUSIONS

Thanks to the in air characterization of the LoS as function of the Wall's gradient, the proportional constant K has been measured and expressed with its uncertainty.

From the data analysis collected during OHU STM thermal balance/cycle test, the error due to the PT1000 position representativeness wrt average of Walls temperature has been derived and considered for the uncertainty estimation.

As result, when OHU will be in flight, the scientific team can use the telemetry of the two PT1000 sensors embedded in OHU structure (at Baffle Wall and Optical Wall location) to have a real time indirect measurement of the LoS and the AKE.

REFERENCES

- [1] Grasset, O., Dougherty, O., Coustenis, A., Bunce, E.J., Erd, C., Titov, D., Blanc, M., Coates, A., Drossart, P., Fletcher, L.N., Hussmann, H., Jaumann, R., Krupp, N., Lebreton, J.-P., Prieto-Ballesteros, O., Tortora, P., Tosi, F., Van Hoolst, T., "Jupiter ICy moons Explorer (JUICE): An ESA mission to orbit Ganymede and to characterise the Jupiter system," PSS 78, 1-21 (2013).
- [2] A. Turella, V. Della Corte, et al., "Structural-Thermal-Optical-Performance (STOP) analysis for the prediction of the line-of-sight stability of JANUS camera on board JUICE ESA mission, " Proc. SPIE 11103, (2019).
- [3] V. Della Corte, G. Noci, A. Turella, R. Paolinetti, et al., "Scientific objectives of JANUS Instrument onboard JUICE mission and key technical solutions for its Optical Head," 2019 IEEE 5th International Workshop on Metrology for AeroSpace (MetroAeroSpace).
- [4] S. Kline, F. McClintock, "Describing Uncertainties in Single-Sample Experiments," Mechanical Engineering, Vol. 75, 1953.